

Spring 2026

Problem 1 (Sp22, Sp26). Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is a continuous function $\int_0^1 x^n f(x) dx = 0$ for all integers $n \geq 1$. Prove that f is identically 0.

Problem 2 (Sp26). Let $\{f_n\}$ be a sequence of differentiable functions on $[0, 1]$ with $f_n(0) = 0$ for all n . Suppose that $\{f'_n\}$ converges uniformly on $[0, 1]$ to a continuous function g .

1. Prove that $\{f_n\}$ converges uniformly on $[0, 1]$ to a function f .
2. Show that f is C^1 on $[0, 1]$, that $f(0) = 0$, and that $f' = g$.

Problem 3 (Sp26). Let f be an entire function such that

$$|f(z)| \leq A + B|z|^m$$

for some constants $A, B \geq 0$ and a nonnegative integer m .

1. Show that f is a polynomial of degree at most m .
2. Deduce that if $|f(z)| \leq C(1 + |z|^{1/2})$ for some $C > 0$, then f is constant.

Problem 4 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26). Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \pi \alpha}$$

What restrictions must be placed on α ?

Problem 5 (Sp26). Let A be a real $n \times n$ matrix such that $A^3 = A$. Describe all possible Jordan canonical forms of A .

Problem 6 (Sp26). Orthogonally diagonalize the symmetric matrix

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

Specifically:

1. Find the eigenvalues and eigenspaces of A .
2. Within the eigenspace of the repeated eigenvalue, use the *Gram–Schmidt process* to produce an orthonormal basis.
3. Form an orthogonal matrix Q from your orthonormal eigenvectors and verify that $Q^T A Q$ is diagonal. Give Q and $Q^T A Q$ explicitly.

Problem 7 (Fa97, Sp09, Sp16, Sp26). Let G be a finite group.

1. If G is non-abelian, show that $G/Z(G)$ is not cyclic, where $Z(G)$ is the center of G .
2. If $|G| = p^n$, with p prime and $n > 0$, show that $Z(G)$ is not trivial and deduce that G has a normal subgroup of order p .
3. If $|G| = p^2$, show that G is abelian.

Problem 8 (Fa83, Sp24, Sp26). 1. Let R be a finite integral domain, i.e., a finite commutative ring with 1 and no zero divisors. Prove that R is a field.

2. Show that if p is the characteristic of R then the Frobenius map $F : R \rightarrow R$, $F(x) = x^p$ is an automorphism (and in particular is bijective).

Problem 9 (Sp26). Let c_n be the number of ways of giving n cents change using coins of values 1, 5, 10 and 25 cents. Find $\lim_{n \rightarrow \infty} c_n/n^3$.

Problem 10 (Sp11, Sp26). The Catalan numbers C_n satisfy $C_0 = 1$, $C_n = C_0 C_{n-1} + C_1 C_{n-2} + \cdots + C_{n-1} C_0$ for $n > 0$. Find the function $\sum_{n=0}^{\infty} C_n x^n$ and use this to evaluate C_n .

Problem 11 (Sp77, Fa81, Sp82, Sp26). Evaluate $\int_{-\infty}^{\infty} \frac{\cos nx}{x^4 + 1} dx$.

Problem 12 (Fa10, Sp21, Sp26). If f is an analytic function from the unit disc into itself with $f(0) = 0$, prove that $|f'(0)| \leq 1$ and find all functions f for which equality holds.

Problem 13 (Sp26). Let $M = (m_{i,j})$, $0 \leq i, j < n$ be an $n \times n$ complex matrix with $m_{i,j} = m_{i+j}$ for constants m_i satisfying $m_i = m_{n+i}$. Find the eigenvalues of M .

Problem 14 (Sp26). For complex square matrices A_i , put

$$F_n(A_1, \dots, A_n) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) A_{\sigma(1)} A_{\sigma(2)} \cdots A_{\sigma(n)}$$

The sum is over the symmetric group S_n , and $\text{sgn}(\sigma)$ is 1 or -1 if σ is an even or odd permutation.

Show that F_n vanishes if all its arguments are $m \times m$ matrices and $n > m^2$.

Problem 15 (Sp26). 1. Let G be a finite group and $\varphi : G \rightarrow \text{GL}_n(\mathbb{C})$ a group homomorphism. Show that $\ker \varphi = \{g \in G \mid \text{tr } \varphi(g) = n\}$.

2. Show that this fails for infinite G .

Problem 16 (Fa11, Sp15, Sp26). Let p be a prime number and G be a group such that $g^p = 1$ for all $g \in G$. Show that if $p = 2$ then G is abelian, and give an example with $p > 2$ where G is not abelian.